

Simulation and Analysis of a Closed N-body Gravitational System

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Introduction

In this report, we consider an N-body gravitational simulator where we use the forward Euler, Leapfrog and fourth-order Runge-Kutta methods to numerically integrate the equations of motion and investigate the conservation of energy, linear momentum and angular momentum. In addition, we examine the convergence, long-term accuracy and computational cost of each numerical method using two- and three-body test cases. We also explore the mathematical theory behind why the Leapfrog method's energy error over time is bounded.

Conserved Quantities and Numerical Methods

A massive body in the presence of another massive body experiences an attractive gravitational force. In a system of massive bodies with masses $m_1, \dots, m_N$ and position vectors $\mathbf{r}_1, \dots, \mathbf{r}_N$ relative to some coordinate origin, the attractive force experienced by one body, say $\mathbf{F}_i$, may be calculated via

$$\mathbf{F}_i = G \sum_{j \neq i}^N \frac{m_i m_j}{|\mathbf{r}_j - \mathbf{r}_i|^2} \cdot \frac{\mathbf{r}_j - \mathbf{r}_i}{|\mathbf{r}_j - \mathbf{r}_i|} \iff \ddot{\mathbf{r}}_i = \mathbf{a}_i = G \sum_{j \neq i}^N m_j \frac{\mathbf{r}_j - \mathbf{r}_i}{|\mathbf{r}_j - \mathbf{r}_i|^3}. \quad (1)$$

For a closed N-body gravitational system, total energy, linear momentum and angular momentum are conserved. Thus, these quantities provide useful diagnostics for assessing the numerical methods used in this report. For the exact dynamics of the isolated system, conservation implies that the time derivative of each quantity is zero.

Throughout this report, a normalised system of units is adopted where Newton's gravitational constant $G = 1$ and one numerical time unit is identified with one second for consistency when presenting simulations.

Conservation of Energy

The total energy in the closed system is given by

$$\underbrace{E}_{\text{Total Energy}} = \underbrace{K}_{\text{Kinetic Energy}} + \underbrace{V}_{\text{Potential Energy}}. \quad (2)$$

Using the kinetic energy formula for N bodies, the total kinetic energy is given by

$$K = \frac{1}{2} \sum_{i=1}^N m_i |\mathbf{v}_i|^2, \quad (3)$$

and the potential energy is given by

$$V = -G \sum_{i < j} \frac{m_i m_j}{|\mathbf{r}_j - \mathbf{r}_i|}, \quad (4)$$

therefore, the total energy in the closed system is given by

$$E = \frac{1}{2} \sum_{i=1}^N m_i |\mathbf{v}_i|^2 - G \sum_{i < j} \frac{m_i m_j}{|\mathbf{r}_j - \mathbf{r}_i|}. \quad (5)$$

Conservation of Linear Momentum

The total linear momentum in the closed system is given by

$$\mathbf{P} = \sum_{i=1}^N \mathbf{p}_i = \sum_{i=1}^N m_i \mathbf{v}_i \quad (6)$$

Conservation of Angular Momentum

The total angular momentum in the closed system is given by

$$\mathbf{L} = \sum_{i=1}^N \mathbf{r}_i \times \mathbf{p}_i = \sum_{i=1}^N m_i \mathbf{r}_i \times \mathbf{v}_i \quad (7)$$

where the symbol $\times$ represents the operation of the cross product.

Forward Euler Method

For an N-body system in three-dimensional space, we start with the initial position vectors $\mathbf{r}_0^1, \mathbf{r}_0^2, \dots, \mathbf{r}_0^N$ and initial velocity vectors $\mathbf{v}_0^1, \mathbf{v}_0^2, \dots, \mathbf{v}_0^N$ of the massive bodies. For simplicity, say we have two massive bodies in a closed system where the first and second massive bodies have the initial position and initial velocity pairs $(\mathbf{r}_0^1, \mathbf{v}_0^1)$ and $(\mathbf{r}_0^2, \mathbf{v}_0^2)$, respectively. Using equation (1), we can calculate the accelerations, $\mathbf{a}_0^1$ and $\mathbf{a}_0^2$, experienced by the two massive bodies. We then advance the position and velocity vectors by one time step, Δt , using the acceleration calculated previously such that

$$(\mathbf{r}_0^1, \mathbf{v}_0^1) \mapsto (\mathbf{r}_0^1 + \Delta t \cdot \mathbf{v}_0^1, \mathbf{v}_0^1 + \Delta t \cdot \mathbf{a}_0^1) = (\mathbf{r}_1^1, \mathbf{v}_1^1) \quad (8)$$

The forward Euler update can be motivated using the definition of the derivative, where the definition of the derivative is given by

$$\frac{dy}{dx} = \lim_{\Delta x \rightarrow 0} \frac{y(x + \Delta x) - y(x)}{\Delta x} \quad (9)$$

For sufficiently small finite Δx , the derivative may be approximated using the forward difference

$$\frac{dy}{dx} \approx \frac{y(x + \Delta x) - y(x)}{\Delta x} \quad (10)$$

If we set $y = \mathbf{r}(t)$ and $x = t$, we find

$$\frac{d\mathbf{r}}{dt} \approx \frac{\mathbf{r}(t + \Delta t) - \mathbf{r}(t)}{\Delta t} \quad (11)$$

$$\iff \mathbf{r}(t + \Delta t) \approx \mathbf{r}(t) + \Delta t \cdot \frac{d\mathbf{r}}{dt} \quad (12)$$

$$\iff \mathbf{r}(t + \Delta t) \approx \mathbf{r}(t) + \Delta t \cdot \mathbf{v}(t) \quad (13)$$

Similarly, if we set $y = \mathbf{v}(t)$ and $x = t$, we find

$$\mathbf{v}(t + \Delta t) \approx \mathbf{v}(t) + \Delta t \cdot \mathbf{a}(t) \quad (14)$$

Applying these updates to each body gives the Forward Euler approximation for an N-body system:

$$\mathbf{r}(t + \Delta t) \approx \mathbf{r}(t) + \Delta t \cdot \mathbf{v}(t) \quad (15)$$

$$\mathbf{v}(t + \Delta t) \approx \mathbf{v}(t) + \Delta t \cdot \mathbf{a}(t) \quad (16)$$

Using the subscript notation to correspond to the n th time step, the $(n + 1)$ th time step of the i th body may be found via

$$\mathbf{r}_{i,n+1} = \mathbf{r}_{i,n} + \Delta t \cdot \mathbf{v}_{i,n} \quad (17)$$

$$\mathbf{v}_{i,n+1} = \mathbf{v}_{i,n} + \Delta t \cdot \mathbf{a}_{i,n} \quad (18)$$

Runge-Kutta Method

For this particular investigation, we will use the fourth-order Runge-Kutta method, often dubbed the RK4 method. Let $\mathbf{r}$ and $\mathbf{v}$ be the stacked position and velocity vectors of all N bodies, and define the system state by $\mathbf{y} = (\mathbf{r}, \mathbf{v})^T$ such that $\dot{\mathbf{y}} = (\mathbf{v}, \mathbf{a}(\mathbf{r}))^T$. Then, we will define the following set of equations

$$\mathbf{k}_1 = \frac{d\mathbf{y}}{dt}(t_n, \mathbf{y}_n), \quad (19)$$

$$\mathbf{k}_2 = \frac{d\mathbf{y}}{dt} \left(t_n + \frac{\Delta t}{2}, \mathbf{y}_n + \frac{\Delta t}{2} \mathbf{k}_1 \right), \quad (20)$$

$$\mathbf{k}_3 = \frac{d\mathbf{y}}{dt} \left(t_n + \frac{\Delta t}{2}, \mathbf{y}_n + \frac{\Delta t}{2} \mathbf{k}_2 \right), \quad (21)$$

$$\mathbf{k}_4 = \frac{d\mathbf{y}}{dt}(t_n + \Delta t, \mathbf{y}_n + \Delta t \mathbf{k}_3). \quad (22)$$

These equations approximate a time step in the dynamics given by

$$(t_{n+1}, \mathbf{y}_{n+1}) = \left(t_n + \Delta t, \mathbf{y}_n + \frac{\Delta t}{6} (\mathbf{k}_1 + 2\mathbf{k}_2 + 2\mathbf{k}_3 + \mathbf{k}_4) \right) \quad (23)$$

The first equation, $\mathbf{k}_1$, evaluates the derivative at the beginning of the time step. This is used to estimate the state at the midpoint, $t + \Delta t/2$, where a second derivative, $\mathbf{k}_2$, is evaluated. A further midpoint estimate gives $\mathbf{k}_3$, and $\mathbf{k}_4$ evaluates the derivative at an estimated state at the end of the step. We then combine these four stages using the weighted average in (23).

Leapfrog Method

The Leapfrog method is a method that is particularly well-suited to long-time integration of Hamiltonian systems because of its symplectic structure. The kick-drift-kick leapfrog (velocity-verlet) method that we have used in this report works as follows: we first perform a half kick on the velocity such that

$$\mathbf{v}_{n+1/2} = \mathbf{v}_n + \frac{\Delta t}{2} \mathbf{a}(\mathbf{r}_n); \quad (24)$$

afterwards, we perform a drift on the position vector by one time step, Δt , where

$$\mathbf{r}_{n+1} = \mathbf{r}_n + \Delta t \cdot \mathbf{v}_{n+1/2}; \quad (25)$$

once we have drifted the position vector, we calculate the acceleration vector at the new position using (1). This new acceleration vector is then used to perform another half-kick to the velocity, where we get

$$\mathbf{v}_{n+1} = \mathbf{v}_{n+1/2} + \frac{\Delta t}{2} \mathbf{a}(\mathbf{r}_{n+1}). \quad (26)$$

This method's favourable long-term conservation properties are attributed to its symplecticity. Here, symplectic refers to a fundamental geometric property of a transformation on phase space. In Hamiltonian mechanics, a state is represented by its position and momenta $(\mathbf{q}, \mathbf{p})$, and the dynamics preserve a geometric structure called the symplectic form. For an N-body system, let $(\mathbf{q}, \mathbf{p}) = (\mathbf{q}_1, \dots, \mathbf{q}_N, \mathbf{p}_1, \dots, \mathbf{p}_N)$ denote the stacked $3N$ -dimensional position and momentum vectors. The symplectic structure, in canonical coordinates, is represented by

$$\Omega = \begin{pmatrix} 0 & I_{3N} \\ -I_{3N} & 0 \end{pmatrix} \quad (27)$$

where I_{3N} is the $3N$ -dimensional identity matrix and the 0's are $3N$ -dimensional block matrices composed entirely of zeros. A map with Jacobian M is symplectic if

$$M^T \Omega M = \Omega. \quad (28)$$

For our N-body system, the Hamiltonian is

$$H(\mathbf{q}, \mathbf{p}) = K(\mathbf{p}) + V(\mathbf{q}) \quad (29)$$

$$= \sum_{i=1}^N \frac{\|\mathbf{p}_i\|^2}{2m_i} - G \sum_{i < j} \frac{m_i m_j}{\|\mathbf{q}_i - \mathbf{q}_j\|} \quad (30)$$

where $K(\mathbf{p})$ is the kinetic energy depending only on momentum whereas $V(\mathbf{q})$ is the potential energy depending only on position. Therefore, this Hamiltonian is separable and allows us to split the evolution into the kick and drift steps. We will now prove that the Hamiltonian above reproduces the equations of motion seen in the code. Using Hamilton's equations, we find

$$\dot{\mathbf{q}}_i = \frac{\partial H}{\partial \mathbf{p}_i} = \frac{\partial K}{\partial \mathbf{p}_i} = \frac{\partial}{\partial \mathbf{p}_i} \frac{\mathbf{p}_i^2}{2m_i} = \frac{\mathbf{p}_i}{m_i} = \mathbf{v}_i \quad (31)$$

and

$$\dot{\mathbf{p}}_i = -\frac{\partial H}{\partial \mathbf{q}_i} = -\frac{\partial V}{\partial \mathbf{q}_i} = -\nabla_i V \quad (32)$$

where we have used ∇_i to represent the partial derivative of the potential with respect to position $\mathbf{q}_i$. Therefore, we find that the rate of change of position is velocity, and the rate of change of momentum is $-\nabla_i V = \mathbf{F}_i$, which is physically equivalent to force. Since $\mathbf{p}_i = m_i \mathbf{v}_i$, we have

$$\dot{\mathbf{p}}_i = m_i \dot{\mathbf{v}}_i = \mathbf{F}_i = m_i \mathbf{a}_i \iff \dot{\mathbf{v}}_i = \mathbf{a}_i \quad (33)$$

Hence, the rate of change of velocity is acceleration. Looking back at our calculations, we may represent our acceleration as

$$\mathbf{a}_i = -\frac{1}{m_i} \nabla_i V \quad (34)$$

For the kinetic part of our Hamiltonian, we have proved that

$$\dot{\mathbf{q}} = \frac{\partial K}{\partial \mathbf{p}} = \mathcal{M}^{-1} \mathbf{p}, \quad \dot{\mathbf{p}} = -\frac{\partial K}{\partial \mathbf{q}} = 0 \quad (35)$$

where $\mathcal{M} = \text{diag}(m_1 I_3, \dots, m_N I_3)$ such that $\mathcal{M}^{-1} = \text{diag}(m_1^{-1} I_3, \dots, m_N^{-1} I_3)$. Solving this system of equations, we find that momentum remains constant and position evolves linearly such that

$$\mathbf{q}' = \mathbf{q} + t \mathcal{M}^{-1} \mathbf{p} \quad (36)$$

$$\mathbf{p}' = \mathbf{p} \quad (37)$$

For some time $t = \Delta t$, we recover the formula for the drift used in our code

$$\mathbf{q}' = \mathbf{q} + \Delta t \cdot \mathbf{v} \quad (38)$$

$$\mathbf{p}' = \mathbf{p} \quad (39)$$

For the potential part of our Hamiltonian, we have proved that

$$\dot{\mathbf{q}} = \frac{\partial V}{\partial \mathbf{p}} = 0, \quad \dot{\mathbf{p}} = -\frac{\partial V}{\partial \mathbf{q}} = \mathbf{F} \quad (40)$$

Solving this system of equations, we find that position remains constant and momentum evolves linearly such that

$$\mathbf{q}' = \mathbf{q} \quad (41)$$

$$\mathbf{p}' = \mathbf{p} + t \mathbf{F} \quad (42)$$

For some time $t = \Delta t$, we recover the formula for the kick used in our code

$$\mathbf{q}' = \mathbf{q} \quad (43)$$

$$\mathbf{v}' = \mathbf{v} + \Delta t \cdot \mathbf{a} \quad (44)$$

The final step is to prove that the sequence of a half-kick followed by a drift followed by a final half-kick is symplectic [1]. We may represent our drift transformation and kick transformation, respectively, via

$$D(\Delta t) : \begin{cases} \mathbf{q}' = \mathbf{q} + \Delta t \cdot \mathcal{M}^{-1} \mathbf{p} \\ \mathbf{p}' = \mathbf{p} \end{cases} \quad (45)$$

$$K(\Delta t) : \begin{cases} \mathbf{q}' = \mathbf{q} \\ \mathbf{p}' = \mathbf{p} - \Delta t \cdot \nabla V(\mathbf{q}) \end{cases} \quad (46)$$

Thus, we need to prove that $\Phi_{\Delta t} = K(\Delta t/2) \circ D(\Delta t) \circ K(\Delta t/2)$ is symplectic. To do this, we may use the theorem that states the composition of symplectic maps is symplectic. Hence, we only need to prove that both $K(\Delta t/2)$ and $D(\Delta t)$ are symplectic, and the proof follows. We will first prove that the kick is symplectic. The Jacobian of the kick, say $M_{K(\Delta t)}$, is given by

$$M_{K(\Delta t)} = \begin{pmatrix} \frac{\partial \mathbf{q}'}{\partial \mathbf{q}} & \frac{\partial \mathbf{q}'}{\partial \mathbf{p}} \\ \frac{\partial \mathbf{p}'}{\partial \mathbf{q}} & \frac{\partial \mathbf{p}'}{\partial \mathbf{p}} \end{pmatrix} = \begin{pmatrix} \frac{\partial}{\partial \mathbf{q}}(\mathbf{q}) & \frac{\partial}{\partial \mathbf{p}}(\mathbf{q}) \\ \frac{\partial}{\partial \mathbf{q}}(\mathbf{p} - \Delta t \cdot \nabla V(\mathbf{q})) & \frac{\partial}{\partial \mathbf{p}}(\mathbf{p} - \Delta t \cdot \nabla V(\mathbf{q})) \end{pmatrix} = \begin{pmatrix} I & 0 \\ -\Delta t \nabla^2 V(\mathbf{q}) & I \end{pmatrix} \quad (47)$$

where the Hessian of the potential, $\nabla^2 V$, is a symmetric matrix composed of second-order partial derivatives. Observe:

$$M_{K(\Delta t)}^T \Omega M_{K(\Delta t)} = \begin{pmatrix} I & 0 \\ -\Delta t \nabla^2 V(\mathbf{q}) & I \end{pmatrix}^T \begin{pmatrix} 0 & I \\ -I & 0 \end{pmatrix} \begin{pmatrix} I & 0 \\ -\Delta t \nabla^2 V(\mathbf{q}) & I \end{pmatrix} \quad (48)$$

$$= \begin{pmatrix} \Delta t \nabla^2 V & I \\ -I & 0 \end{pmatrix} \begin{pmatrix} I & 0 \\ -\Delta t \nabla^2 V(\mathbf{q}) & I \end{pmatrix} \quad (49)$$

$$= \begin{pmatrix} \Delta t \nabla^2 V - \Delta t \nabla^2 V & I \\ -I & 0 \end{pmatrix} \quad (50)$$

$$= \Omega \quad (51)$$

The Jacobian of the drift, say $M_{D(\Delta t)}$, is given by

$$M_{D(\Delta t)} = \begin{pmatrix} \frac{\partial \mathbf{q}'}{\partial \mathbf{q}} & \frac{\partial \mathbf{q}'}{\partial \mathbf{p}} \\ \frac{\partial \mathbf{p}'}{\partial \mathbf{q}} & \frac{\partial \mathbf{p}'}{\partial \mathbf{p}} \end{pmatrix} = \begin{pmatrix} \frac{\partial}{\partial \mathbf{q}}(\mathbf{q} + \Delta t \cdot \mathcal{M}^{-1} \mathbf{p}) & \frac{\partial}{\partial \mathbf{p}}(\mathbf{q} + \Delta t \cdot \mathcal{M}^{-1} \mathbf{p}) \\ \frac{\partial}{\partial \mathbf{q}}(\mathbf{p}) & \frac{\partial}{\partial \mathbf{p}}(\mathbf{p}) \end{pmatrix} = \begin{pmatrix} I & \Delta t \mathcal{M}^{-1} \\ 0 & I \end{pmatrix} \quad (52)$$

Observe:

$$M_{D(\Delta t)}^T \Omega M_{D(\Delta t)} = \begin{pmatrix} I & \Delta t \mathcal{M}^{-1} \\ 0 & I \end{pmatrix}^T \begin{pmatrix} 0 & I \\ -I & 0 \end{pmatrix} \begin{pmatrix} I & \Delta t \mathcal{M}^{-1} \\ 0 & I \end{pmatrix} \quad (53)$$

$$= \begin{pmatrix} 0 & I \\ -I & \Delta t \mathcal{M}^{-1} \end{pmatrix} \begin{pmatrix} I & \Delta t \mathcal{M}^{-1} \\ 0 & I \end{pmatrix} \quad (54)$$

$$= \begin{pmatrix} 0 & I \\ -I & \Delta t \mathcal{M}^{-1} - \Delta t \mathcal{M}^{-1} \end{pmatrix} \quad (55)$$

$$= \Omega \quad (56)$$

We have thus proven that both $K(\Delta t)$ and $D(\Delta t)$ are symplectic and it follows that $\Phi_{\Delta t} = K(\Delta t/2) \circ D(\Delta t) \circ K(\Delta t/2)$ is symplectic. We may verify this by observing the following:

$$\Phi_{\Delta t}^T \Omega \Phi_{\Delta t} = (M_{K(\Delta t/2)} M_{D(\Delta t)} M_{K(\Delta t/2)})^T \Omega (M_{K(\Delta t/2)} M_{D(\Delta t)} M_{K(\Delta t/2)}) \quad (57)$$

$$= M_{K(\Delta t/2)}^T M_{D(\Delta t)}^T M_{K(\Delta t/2)}^T \Omega M_{K(\Delta t/2)} M_{D(\Delta t)} M_{K(\Delta t/2)} \quad (58)$$

$$= M_{K(\Delta t/2)}^T M_{D(\Delta t)}^T (M_{K(\Delta t/2)}^T \Omega M_{K(\Delta t/2)}) M_{D(\Delta t)} M_{K(\Delta t/2)} \quad (59)$$

$$= M_{K(\Delta t/2)}^T (M_{D(\Delta t)}^T \Omega M_{D(\Delta t)}) M_{K(\Delta t/2)} \quad (60)$$

$$= M_{K(\Delta t/2)}^T \Omega M_{K(\Delta t/2)} \quad (61)$$

$$= \Omega \quad (62)$$

This symplecticity preserves the underlying geometric structure of the Hamiltonian phase space. However, the Leapfrog method does not exactly conserve the original Hamiltonian; for stable bounded trajectories its energy error typically remains bounded and oscillatory. We will show this first by introducing the Liouville operator L_H defined by

$$L_H f \equiv \{f, H\} = \frac{\partial f}{\partial \mathbf{q}} \frac{\partial H}{\partial \mathbf{p}} - \frac{\partial H}{\partial \mathbf{q}} \frac{\partial f}{\partial \mathbf{p}} = \frac{df}{dt} \quad (63)$$

Essentially, the Liouville operator generates the time derivative of a phase-space quantity. From (63), we find

$$L_H f = \frac{df}{dt} \quad (64)$$

$$\implies f(t) = \exp\{t L_H\} f(0) \quad (65)$$

therefore, $\exp\{t L_H\}$ evolves the system forward by time t . Using (63), we find

$$L_H \mathbf{q} = \frac{\partial \mathbf{q}}{\partial \mathbf{q}} \frac{\partial H}{\partial \mathbf{p}} - \frac{\partial H}{\partial \mathbf{q}} \frac{\partial \mathbf{q}}{\partial \mathbf{p}} \quad (66)$$

$$= \frac{\partial \mathbf{q}}{\partial \mathbf{q}} \frac{\partial (K + V)}{\partial \mathbf{p}} - \frac{\partial (K + V)}{\partial \mathbf{q}} \frac{\partial \mathbf{q}}{\partial \mathbf{p}} \quad (67)$$

$$= \frac{\partial \mathbf{q}}{\partial \mathbf{q}} \frac{\partial K}{\partial \mathbf{p}} - \frac{\partial K}{\partial \mathbf{q}} \frac{\partial \mathbf{q}}{\partial \mathbf{p}} + \frac{\partial \mathbf{q}}{\partial \mathbf{q}} \frac{\partial V}{\partial \mathbf{p}} - \frac{\partial V}{\partial \mathbf{q}} \frac{\partial \mathbf{q}}{\partial \mathbf{p}} \quad (68)$$

$$= \{\mathbf{q}, K\} + \{\mathbf{q}, V\} \quad (69)$$

$$= L_K \mathbf{q} + L_V \mathbf{q} \quad (70)$$

However, since $L_V \mathbf{q} = \{\mathbf{q}, V\} = 0$, we find that

$$\dot{\mathbf{q}} = L_H \mathbf{q} = L_K \mathbf{q} = \mathcal{M}^{-1} \mathbf{p} \implies \mathbf{q}(t + \Delta t) = \exp\{\Delta t L_K\} \mathbf{q}(t) = \mathbf{q}(t) + \Delta t \cdot \mathcal{M}^{-1} \mathbf{p} \quad (71)$$

Similarly, we find

$$L_H \mathbf{p} = \frac{\partial \mathbf{p}}{\partial \mathbf{q}} \frac{\partial H}{\partial \mathbf{p}} - \frac{\partial H}{\partial \mathbf{q}} \frac{\partial \mathbf{p}}{\partial \mathbf{p}} \quad (72)$$

$$= \frac{\partial \mathbf{p}}{\partial \mathbf{q}} \frac{\partial (K+V)}{\partial \mathbf{p}} - \frac{\partial (K+V)}{\partial \mathbf{q}} \frac{\partial \mathbf{p}}{\partial \mathbf{p}} \quad (73)$$

$$= \frac{\partial \mathbf{p}}{\partial \mathbf{q}} \frac{\partial K}{\partial \mathbf{p}} - \frac{\partial K}{\partial \mathbf{q}} \frac{\partial \mathbf{p}}{\partial \mathbf{p}} + \frac{\partial \mathbf{p}}{\partial \mathbf{q}} \frac{\partial V}{\partial \mathbf{p}} - \frac{\partial V}{\partial \mathbf{q}} \frac{\partial \mathbf{p}}{\partial \mathbf{p}} \quad (74)$$

$$= \{\mathbf{p}, K\} + \{\mathbf{p}, V\} \quad (75)$$

$$= L_K \mathbf{p} + L_V \mathbf{p} \quad (76)$$

However, since $L_K \mathbf{p} = \{\mathbf{p}, K\} = 0$, we find that

$$\dot{\mathbf{p}} = L_H \mathbf{p} = L_V \mathbf{p} = -\nabla V \implies \mathbf{p}(t + \Delta t) = \exp\{\Delta t L_V\} \mathbf{p}(t) = \mathbf{p}(t) - \Delta t \nabla V \quad (77)$$

Therefore, applying the generator $\exp\{\Delta t L_H\}$ to $(\mathbf{q}, \mathbf{p})$ would produce $(\mathbf{q}(t + \Delta t), \mathbf{p}(t + \Delta t))$, simultaneously updating both the position and momenta. However, our leapfrog method does not apply the full generator to the phase space state but rather applies three generators

$$\Phi_{\Delta t} = \exp\left\{\frac{\Delta t}{2} L_V\right\} \exp\{\Delta t L_K\} \exp\left\{\frac{\Delta t}{2} L_V\right\} \quad (78)$$

The rightmost generator first freezes $\mathbf{q}$ whilst it updates $\mathbf{p}$; the middle generator then freezes $\mathbf{p}$ whilst it updates $\mathbf{q}$; and lastly, the leftmost generator freezes $\mathbf{q}$ again whilst it updates $\mathbf{p}$. The question still remains: why does this transformation not conserve the energy of the system? To address this, we will apply the Baker-Campbell-Hausdorff formula, which states that for two non-commuting variables X and Y , the product of their exponentials is given by

$$\exp\{X\} \exp\{Y\} = \exp\left\{X + Y + \frac{1}{2}[X, Y] + \frac{1}{12}[X, [X, Y]] + \frac{1}{12}[Y, [Y, X]] + \dots\right\} \quad (79)$$

Substituting $X = \Delta t L_V/2$ and $Y = \Delta t L_K$, we find

$$\begin{aligned} \exp\left\{\frac{\Delta t}{2} L_V\right\} \exp\{\Delta t L_K\} &= \exp\left\{\Delta t \left(\frac{1}{2} L_V + L_K\right) \right. \\ &\quad + \Delta t^2 \left(\frac{1}{4} [L_V, L_K]\right) \\ &\quad + \Delta t^3 \left(\frac{1}{48} [L_V, [L_V, L_K]] + \frac{1}{24} [L_K, [L_K, L_V]]\right) \\ &\quad \left. + \mathcal{O}(\Delta t^4)\right\} \end{aligned} \quad (80)$$

For bookkeeping, we shall now call

$$Z = \Delta t \left(\frac{1}{2} L_V + L_K\right) + \Delta t^2 \left(\frac{1}{4} [L_V, L_K]\right) + \Delta t^3 \left(\frac{1}{48} [L_V, [L_V, L_K]] + \frac{1}{24} [L_K, [L_K, L_V]]\right) + \mathcal{O}(\Delta t^4) \quad (81)$$

We will now apply the Baker-Campbell-Hausdorff formula again for Z and $X = \Delta t L_V/2$, given by

$$\Phi_{\Delta t} = \exp\{Z\} \exp\{X\} \quad (82)$$

$$= \exp\left\{Z + X + \frac{1}{2}[Z, X] + \frac{1}{12}[Z, [Z, X]] + \frac{1}{12}[X, [X, Z]] + \dots\right\} \quad (83)$$

From (83), we find

$$\begin{aligned}
Z + X &= \Delta t \left(\frac{1}{2} L_V + L_K \right) \\
&+ \Delta t^2 \left(\frac{1}{4} [L_V, L_K] \right) \\
&+ \Delta t^3 \left(\frac{1}{48} [L_V, [L_V, L_K]] + \frac{1}{24} [L_K, [L_K, L_V]] \right) \\
&+ \mathcal{O}(\Delta t^4)
\end{aligned} \tag{84}$$

$$\begin{aligned}
\frac{1}{2} [Z, X] &= \Delta t^2 \left(\frac{1}{4} [L_K, L_V] \right) \\
&+ \Delta t^3 \left(\frac{1}{16} [[L_V, L_K], L_V] \right) \\
&+ \mathcal{O}(\Delta t^4)
\end{aligned} \tag{85}$$

$$\begin{aligned}
\frac{1}{12} [Z, [Z, X]] &= \Delta t^3 \left(\frac{1}{48} [L_V, [L_K, L_V]] + \frac{1}{24} [L_K, [L_K, L_V]] \right) \\
&+ \mathcal{O}(\Delta t^4)
\end{aligned} \tag{86}$$

$$\begin{aligned}
\frac{1}{12} [X, [X, Z]] &= \Delta t^3 \left(-\frac{1}{48} [L_V, [L_K, L_V]] \right) \\
&+ \mathcal{O}(\Delta t^4)
\end{aligned} \tag{87}$$

Using commutation rules and substituting these identities into (83), we find

$$\Phi_{\Delta t} = \exp \left\{ \Delta t (L_K + L_V) + \Delta t^3 \left(\frac{1}{12} [L_K, [L_K, L_V]] - \frac{1}{24} [L_V, [L_V, L_K]] \right) + \mathcal{O}(\Delta t^5) \right\} \tag{88}$$

$$= \exp \left\{ \Delta t \left[L_K + L_V + \Delta t^2 \left(\frac{1}{12} [L_K, [L_K, L_V]] - \frac{1}{24} [L_V, [L_V, L_K]] \right) + \mathcal{O}(\Delta t^4) \right] \right\} \tag{89}$$

$$= \exp \{ \Delta t L_{\tilde{H}} \} \tag{90}$$

where

$$\tilde{H} = H + \Delta t^2 H' + \mathcal{O}(\Delta t^4) \tag{91}$$

Hence, leapfrog doesn't follow the true Hamiltonian H . Our leapfrog numerical trajectory may be interpreted as following the dynamics generated by the modified Hamiltonian $\tilde{H}^{(2)} = H + \Delta t^2 H'$. The modified Hamiltonian seen in (91) contains higher-order terms; however, for the qualitative analysis here, we retain terms through $\mathcal{O}(\Delta t^2)$ in the modified Hamiltonian. In an attempt to understand why the relative energy error is oscillatory for the leapfrog method, we will assume that, to the order retained in our Baker-Campbell-Hausdorff calculation, our leapfrog method is modelled as exact Hamiltonian evolution under $\tilde{H}^{(2)} = H + \Delta t^2 H'$. Within this truncated modified-Hamiltonian approximation, $\tilde{H}^{(2)}$ is treated as conserved, such that

$$\frac{d\tilde{H}^{(2)}}{dt} = \{ \tilde{H}^{(2)}, \tilde{H}^{(2)} \} = 0. \tag{92}$$

Now, the Hamiltonian $H'(\mathbf{q}, \mathbf{p})$ acts as a correction function on phase space composed of nested Poisson brackets and is a function of $\mathbf{q}$ and $\mathbf{p}$. For stable, bounded periodic or quasiperiodic orbits, both $\mathbf{q}$ and $\mathbf{p}$ repeatedly follow a bounded region of phase space, and thus $H'(\mathbf{q}, \mathbf{p})$ will vary in a bounded and recurring manner. Following our assumption, if $\tilde{H}^{(2)}$ is constant and $\Delta t^2 H'$ varies, H must compensate equally and oppositely to the correction function. This oscillation in the Hamiltonian H is what causes the oscillations seen in our relative energy error.

Analysis

In this section, we analyse the results obtained from our numerical simulations to compare the strengths and limitations of each numerical method, including their ability to conserve energy, linear momentum and angular momentum. We will also conduct an investigate their long-term stability and computational efficiency.

Validation of Orbital Trajectories

To assess the validity of our models, we may compare the fourth-order Runge-Kutta solution with a known circular two-body solution. Suppose that for any $\theta \in [0, 2\pi)$, we initially have two bodies with equal mass $m_1 = m_2 = 1$ and the following position and velocity vectors:

$$\begin{aligned} \mathbf{r}_1 &= (\cos \theta, \sin \theta, 0), \quad \mathbf{v}_1 = \left(-\frac{1}{2} \sin \theta, \frac{1}{2} \cos \theta, 0\right), \\ \mathbf{r}_2 &= (-\cos \theta, -\sin \theta, 0), \quad \mathbf{v}_2 = \left(\frac{1}{2} \sin \theta, -\frac{1}{2} \cos \theta, 0\right). \end{aligned} \quad (93)$$

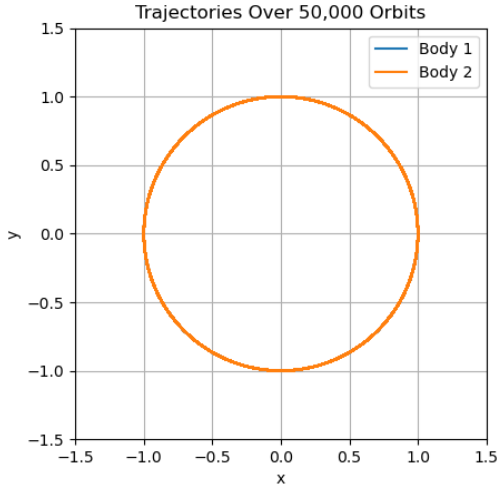
Calculating the gravitational acceleration for each body using (1), we find

$$\mathbf{a}_1 = \left(-\frac{1}{4} \cos \theta, -\frac{1}{4} \sin \theta, 0\right), \quad \mathbf{a}_2 = \left(\frac{1}{4} \cos \theta, \frac{1}{4} \sin \theta, 0\right). \quad (94)$$

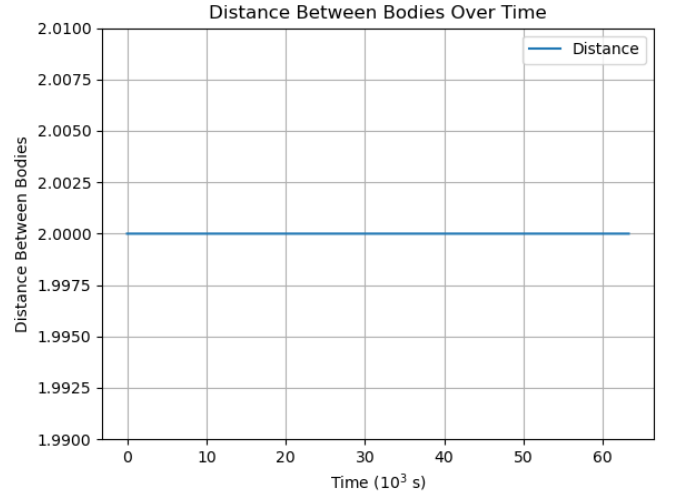
These initial conditions satisfy the acceleration equation for circular motion, given by

$$\|\mathbf{a}\| = \frac{\|\mathbf{v}\|^2}{\|\mathbf{r}\|} \quad (95)$$

Therefore, under the exact Newtonian dynamics, these initial conditions produce circular motion with period 4π . Agreement with this known solution over many orbital periods therefore provides a useful validation test for the numerical implementation.



(a) Numerical trajectories of both massive bodies over 50,000 orbital periods.



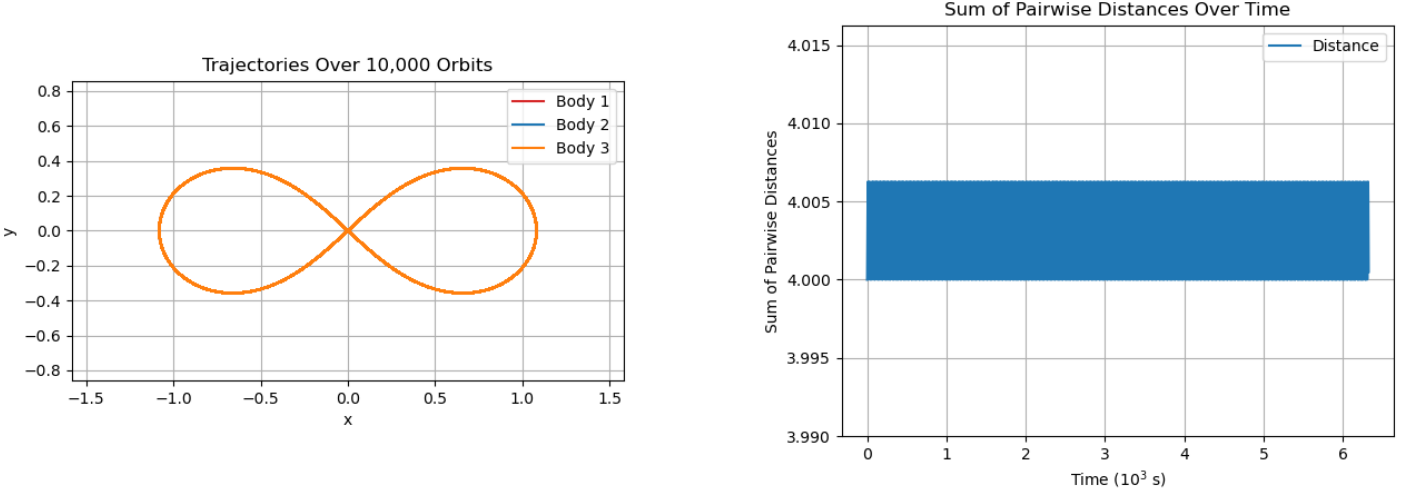
(b) Distance between both massive bodies over 50,000 orbital periods.

Figure 1: Numerical validation of a massive two-body system using the fourth-order Runge-Kutta method with $\Delta t = 0.1$. The trajectories retain the expected circular geometry, whilst the distance between the massive bodies remains approximately constant.

Now that we have assessed that our simulation is accurate over many simple two-body system orbital periods, we may introduce, and use throughout the remainder of this section, a more complex three-body system with $m_1 = m_2 = m_3 = 1$ given by the following initial conditions [2]:

$$\begin{aligned} \mathbf{r}_1 &= (-0.97000436, 0.24308753, 0), \quad \mathbf{v}_1 = (0.466203685, 0.432365730, 0), \\ \mathbf{r}_2 &= (0.97000436, -0.24308753, 0), \quad \mathbf{v}_2 = (0.466203685, 0.432365730, 0), \\ \mathbf{r}_3 &= (0, 0, 0), \quad \mathbf{v}_3 = (-0.932407370, -0.864731460, 0). \end{aligned} \quad (96)$$

These initial conditions correspond to the known figure-eight periodic orbit shown in Figure 2, where we also assess the sum of pairwise distances between the three massive bodies.



(a) Numerical trajectories of the three massive bodies over 10,000 orbital periods.

(b) The sum of pairwise distances between the three massive bodies over 10,000 orbital periods.

Figure 2: Numerical simulation of a massive three-body system using the fourth-order Runge-Kutta method with $\Delta t = 0.1$. The numerical trajectories retain the characteristic figure-eight geometry over the simulated interval. Furthermore, the sum of pairwise distances remains bounded and oscillatory.

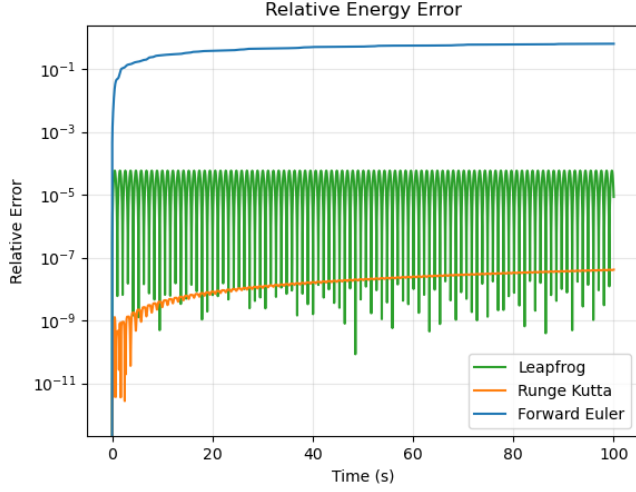
Conservation Properties

In this section, we will evaluate how well the forward Euler, Runge-Kutta, and Leapfrog methods preserve conserved quantities such as energy, linear momentum and angular momentum over many orbital periods. To aid us in our study, we will introduce the dimensionless relative error measure calculated by dividing the absolute error by the absolute initial value of the conserved quantity. For example, for energy, the relative energy error is given by

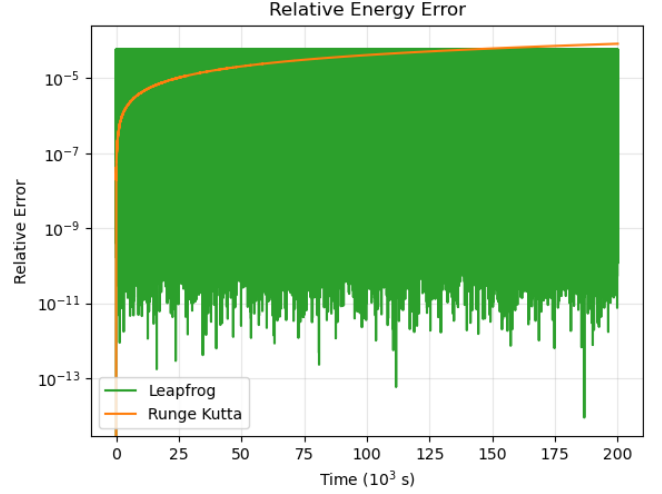
$$\epsilon(E) = \frac{|E - E_0|}{|E_0|} \quad (97)$$

A relative error of 0.05 indicates a 5% deviation from the initial energy, which, under exact dynamics, would remain constant. In the following analysis, the energy error is measured in relative error, whereas linear momentum and angular momentum are measured using absolute error due to their near-zero initial linear and angular momentum, respectively. Since linear momentum, $\mathbf{P}$, and angular momentum, $\mathbf{L}$, are represented by vectors, the absolute error is given by

$$\epsilon(\mathbf{P}) = \|\mathbf{P} - \mathbf{P}_0\|, \quad \epsilon(\mathbf{L}) = \|\mathbf{L} - \mathbf{L}_0\|. \quad (98)$$



(a) Numerical simulation of the conservation of energy using the forward Euler, Runge-Kutta, and Leapfrog methods with $\Delta t = 0.01$ for $t = 100$ seconds.



(b) Numerical simulation of the conservation of energy using the Runge-Kutta and Leapfrog methods with $\Delta t = 0.01$ for $t = 200,000$ seconds.

Figure 3: The forward Euler method is inaccurate at conserving energy over long periods. However, the fourth-order Runge-Kutta method initially showcases the smallest relative energy error, while at sufficiently long times the bounded error of the Leapfrog method becomes smaller than the secularly increasing Runge-Kutta energy error.

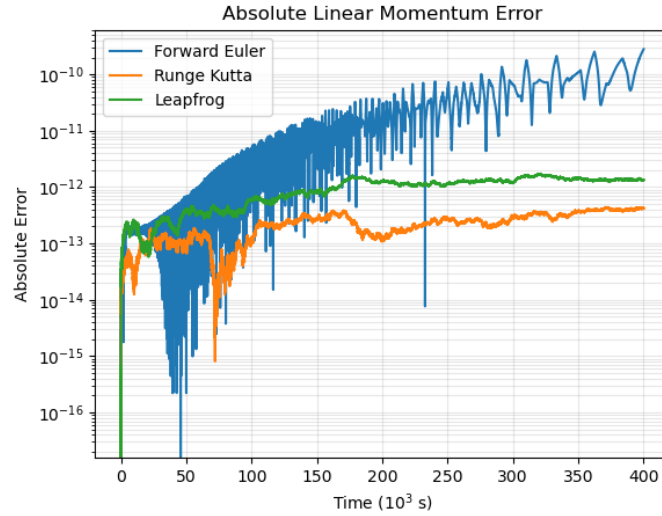
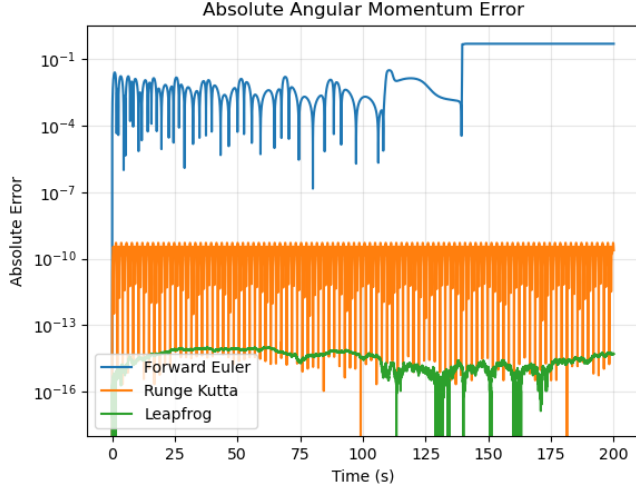
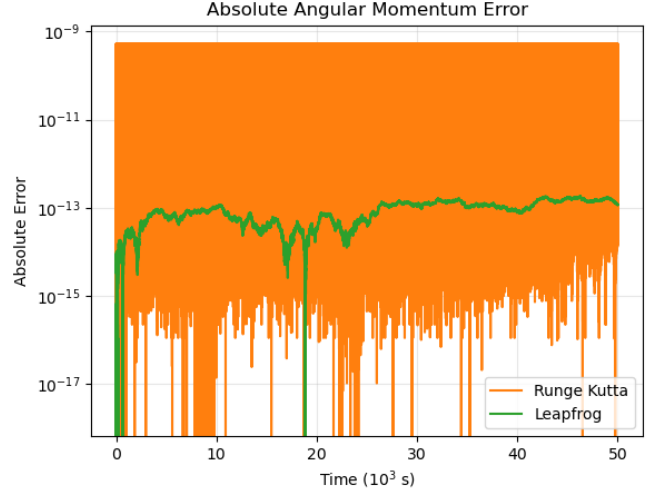


Figure 4: Numerical simulation of the conservation of linear momentum using the forward Euler, Runge-Kutta, and Leapfrog methods with $\Delta t = 0.01$ for $t = 400,000$ seconds. The fourth-order Runge-Kutta and Leapfrog methods remain at an error of order 10^{-13} – 10^{-12} , whereas the forward Euler method exhibits greater numerical error accumulation over the interval shown.



(a) Numerical simulation of the conservation of angular momentum using the forward Euler, Runge-Kutta, and Leapfrog methods with $\Delta t = 0.01$ for $t = 200$ seconds.



(b) Numerical simulation of the conservation of angular momentum using the Runge-Kutta and Leapfrog methods with $\Delta t = 0.01$ for $t = 50,000$ seconds.

Figure 5: Initially, the forward Euler method rapidly diverges towards an absolute angular momentum error of order 10^{-1} and then experiences a catastrophic numerical instability at approximately $t = 140$ seconds. The Leapfrog method preserves angular momentum most accurately in this test, with the error remaining close to numerical precision. Lastly, the fourth-order Runge-Kutta method's angular momentum error remains between 10^{-13} and 10^{-9} over the interval shown.

Convergence Analysis

For a numerical method with order p , the global discretisation error is expected to scale asymptotically as $E \propto (\Delta t)^p$ for sufficiently small Δt . Taking the logarithm, we find $\log E = p \log \Delta t + \log C$. Hence, on a log-log plot, we expect the slope to approximately reflect the order of the method. The forward Euler, Runge-Kutta, and Leapfrog methods are first-, fourth-, and second-order methods, respectively [3].

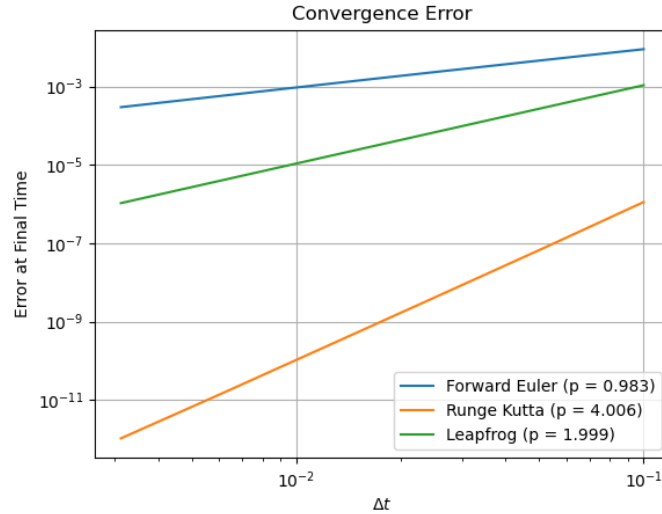


Figure 6: Plot of the convergence error for various values of Δt against a higher-accuracy numerical reference solution using the fourth-order Runge-Kutta method with $\Delta t = 10^{-8}$ measured at $T = 0.1$ seconds. Here, our error, E , denotes the norm of the difference between the simulated and reference position vectors at the fixed final time $t = 0.1$ seconds.

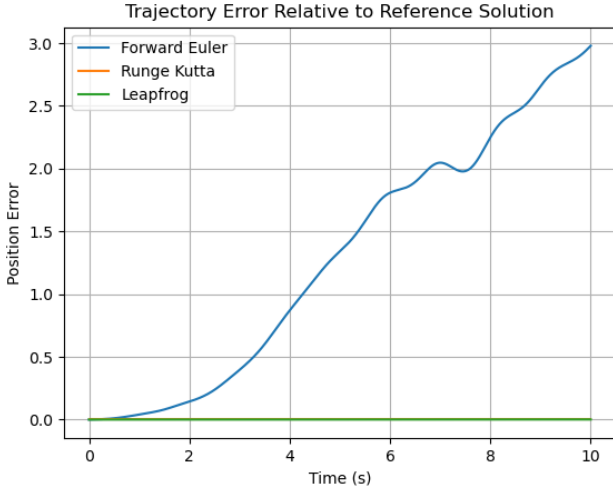
The approximately measured gradients of 0.983, 4.006, and 1.999 agree closely with the expected first-, fourth-, and second-order convergence of the forward Euler, Runge-Kutta and Leapfrog methods, respectively.

Long-Term Trajectory Accuracy

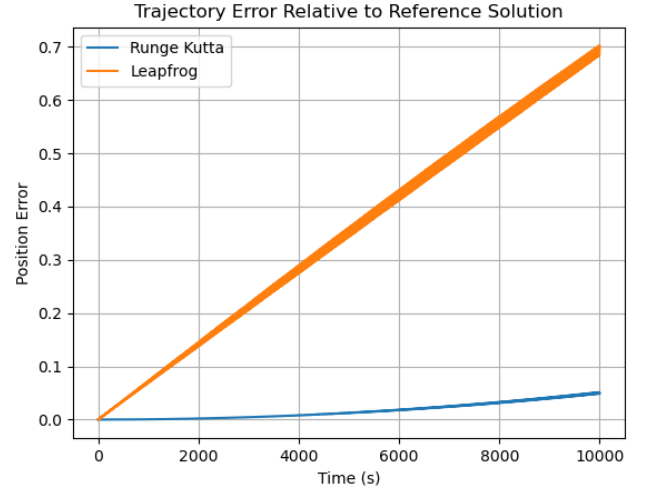
For long-time orbital integrations, it is useful to assess how rapidly numerical trajectories diverge from a higher-accuracy reference solution. Therefore, understanding the long-term stability of the numerical methods used in such a simulator is indispensable. In the following, we evaluate the long-term stability of the forward Euler, Runge-Kutta, and Leapfrog methods using a higher-accuracy numerical reference solution. We will define the position-error metric using the Frobenius norm given by

$$\epsilon_{\mathbf{r}}(t) = \|\mathbf{R}_{\text{sim}}(t) - \mathbf{R}_{\text{ref}}(t)\|_F = \sqrt{\sum_{i=1}^N \|\mathbf{r}_i^{\text{sim}}(t) - \mathbf{r}_i^{\text{ref}}(t)\|^2} \quad (99)$$

where $\mathbf{R}_{\text{sim}}$ and $\mathbf{R}_{\text{ref}}$ are $N \times 3$ matrices composed of the position vectors $\mathbf{r}_i^{\text{sim}}$ and $\mathbf{r}_i^{\text{ref}}$ representing the simulated solution and the reference solution, respectively. Additionally, the subscript F denotes the Frobenius norm operation.



(a) Numerical simulation of the position error of the forward Euler, Runge-Kutta, and Leapfrog methods with $\Delta t = 0.01$ for $t = 10$ seconds.



(b) Numerical simulation of the position error of the Runge-Kutta and Leapfrog methods with $\Delta t = 0.01$ for $t = 10,000$ seconds.

Figure 7: Numerical simulation of the position error of the forward Euler, fourth-order Runge-Kutta, and Leapfrog methods with $\Delta t = 0.01$. The reference simulation is a higher-accuracy numerical approximation to the true solution using the fourth-order Runge-Kutta method with $\Delta t = 10^{-4}$.

Figure 7a confirms that the forward Euler method accumulates a substantial error even after very short periods. For this test case and step size, the forward Euler method accumulates substantial trajectory error over a short interval, making it unsuitable for long-term orbital integration at this resolution. Moreover, Figure 7b supports approximately linear growth in the Leapfrog method position error over the interval investigated. Comparatively, over the interval investigated, the fourth-order Runge-Kutta trajectory remains substantially closer to the higher-accuracy reference solution.

Computational Efficiency

The computational demand of each numerical method is an essential factor to consider when constructing a model to simulate an arbitrary N-body system. To measure the runtime, each method was JIT-compiled before timing to exclude the Numba compilation overhead from impacting the results. Plotting and diagnostic calculations were also excluded. From a heuristic viewpoint, per iteration, we expect the forward Euler method to be the cheapest since it calls the acceleration function once, followed by the Leapfrog method, which calls it twice, and finally the Runge-Kutta method, which calls it four times. This is what our results show. In this construction of an N-body simulation, after performing one million iterations of each method fifty times, we achieved the following runtimes:

Method	Mean Runtime (s)
Forward Euler	3.839 ± 0.016
Runge-Kutta	10.767 ± 0.038
Leapfrog	6.070 ± 0.026

Table 1: Mean runtime and sample standard deviation over fifty simulations of one million iterations.

From the results above, we find that the Runge-Kutta method's runtime is $\sim 180.5\%$ longer than the forward Euler method's runtime and $\sim 77.4\%$ longer than the Leapfrog method's runtime. Over the range tested, the measured runtime scales approximately linearly with the number of iterations.

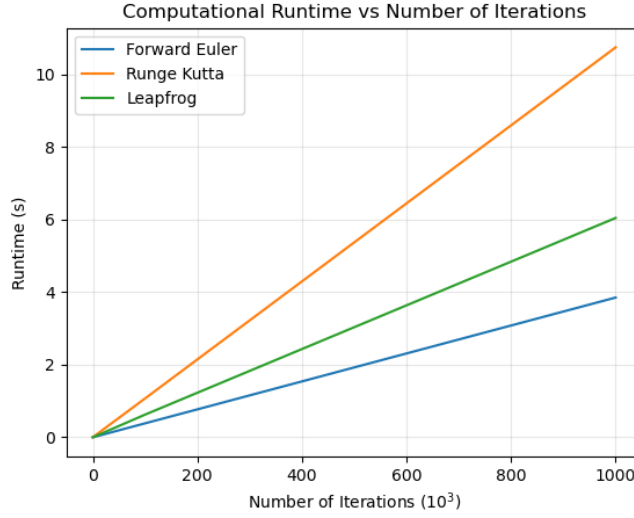
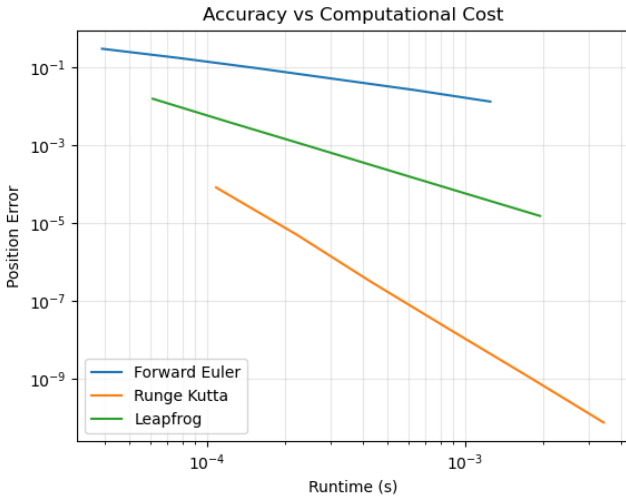
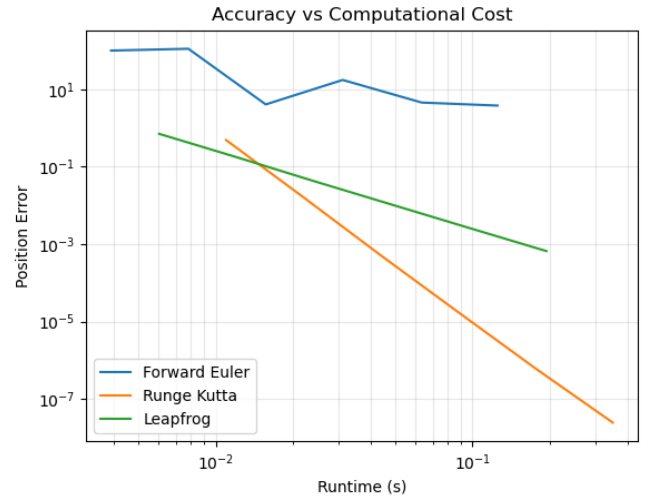


Figure 8: Simulation runtime against the number of numerical method iterations.

Now that we have examined both the long-term stability and runtimes of our three numerical methods, an interesting question remains: which method provides the greatest position accuracy per unit of runtime for different values of Δt ?



(a) Plot of position error against mean computational runtime over one hundred repeated simulations at $t = 1$ second.



(b) Plot of position error against mean computational runtime over one hundred repeated simulations at $t = 100$ seconds.

Figure 9: Position error against mean computational runtime for various values of Δt .

The forward Euler method’s strength is in its relatively quick computation runtime. However, the trade-off for such speed can be seen in its position error. We observe that the forward Euler method can reach one second of simulation time before both the Runge-Kutta and Leapfrog methods.

$\Delta t = 0.1$	Forward Euler	Leapfrog	Runge-Kutta
Runtime (s)	0.0000392	0.0000616	0.0001083
Runtime Ratio to Euler	1.0000000	1.5722271	2.7647900
Error	0.2920214	0.0152455	0.0000811
Error Ratio to RK4	3602.3808	188.06899	1.0000000

Table 2: Computational runtime and error statistics for the three numerical methods.

Observe Figure 9a, for the same simulated interval with $\Delta t = 0.1$ (left-most point on each line), the forward Euler method requires the least computational runtime, where it achieves an error of $\sim 2.9 \times 10^{-1}$ and it does so in $\sim 3.9 \times 10^{-5}$ seconds. However, for the same $\Delta t = 0.1$, the Leapfrog method achieves an error of $\sim 1.6 \times 10^{-2}$ in $\sim 6.2 \times 10^{-5}$ seconds. Thus, for approximately 1.57 times the runtime, we gain approximately 19.15 times smaller position error. A similar argument holds for the Runge-Kutta method as well, where for $\Delta t = 0.1$ it achieved an error of $\sim 8.1 \times 10^{-5}$ in $\sim 1.1 \times 10^{-4}$ seconds. Hence, for approximately 2.76 times the runtime, we gain approximately 3602.38 times smaller position error. Defining an error multiple as

$$R_{\text{Euler}} = \frac{\epsilon_{\text{Euler}}}{\epsilon_{\text{Runge-Kutta}}}, \quad R_{\text{Leapfrog}} = \frac{\epsilon_{\text{Leapfrog}}}{\epsilon_{\text{Runge-Kutta}}}, \quad (100)$$

we may plot the accuracy difference for each value of Δt :

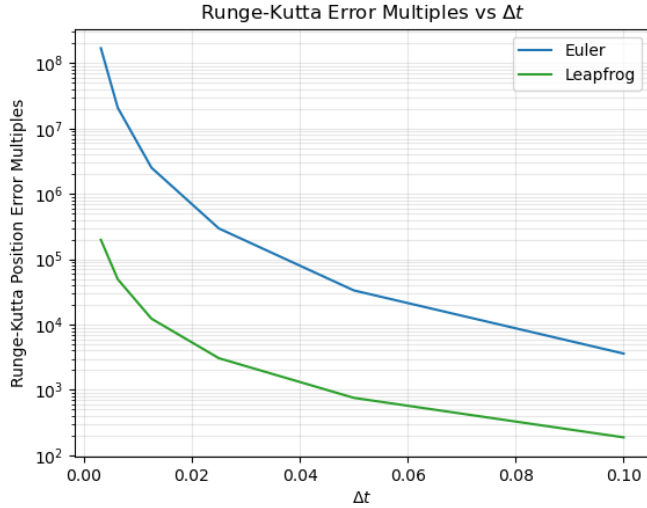


Figure 10: Plot of Runge-Kutta method position error multiples against various values of Δt .

We observe that, over the range of Δt investigated, the error difference between the methods increases as Δt decreases. Previously, from Figure 8, we found that the simulation runtime is approximately linear over the range tested and therefore the simulation runtime for the Leapfrog and Runge-Kutta methods will be approximately 1.57 and 2.76 times longer than the forward Euler method, respectively. Hence, the trade-off for a longer waiting time appears to be a greatly reduced position error. Therefore, we may claim that, out of the three numerical methods discussed and for the cases investigated in this report, despite the longer runtime, Runge-Kutta provides the best positional accuracy. The Runge-Kutta method is used in professional applications; NASA’s GMAT (General Mission Analysis Tool) testing documentation lists several orbit integrators, including the eighth-order Runge-Kutta method [4].

Conclusion

To conclude, having validated the numerical implementation against known test cases, we have assessed the forward Euler, Leapfrog and fourth-order Runge-Kutta methods’ ability to conserve energy, linear momentum and angular momentum. Initially, the Runge-Kutta method delivered the smallest energy error; however, while the Leapfrog method exhibited bounded, oscillatory energy error in the stable orbit investigated, the Leapfrog method ultimately exhibits the smallest energy error. For the Leapfrog and Runge-Kutta methods, the linear momentum and angular momentum remain small at the order of $\sim 10^{-13}$ and $\sim 10^{-10}$, respectively, whereas the forward Euler method exhibits secular growth error, eventually reaching a catastrophic numerical instability.

After the convergence analysis, we found that the measured convergence slopes closely matched the expected first-, fourth-, and second-order behaviour of the forward Euler, Runge-Kutta, and Leapfrog methods, respectively.

The long-term trajectory accuracy showed that, over the tested interval, the Runge-Kutta method was substantially more accurate than the Leapfrog method against a higher-accuracy numerical approximation, while the forward Euler method accumulated much larger trajectory error.

Lastly, we studied the computational efficiency of the three numerical methods displayed in this report. We found that the mean runtimes over fifty simulations of one million iterations were 3.839 ± 0.016 , 6.070 ± 0.026 and 10.767 ± 0.038 for the forward Euler, Leapfrog and Runge-Kutta methods, respectively. When we plotted the position error against the computational runtime for each method, we found that the Runge-Kutta method provided the smallest position error of the three methods for the cases investigated. Integrating over one second of simulation time with $\Delta t = 0.003125$, we found that the Runge-Kutta method had a 1.98×10^5 times smaller position error compared to the Leapfrog method for approximately 1.76 times the computational runtime.

Limitations

The gravitational acceleration function contains a $(\mathbf{r}_j - \mathbf{r}_i)/\|\mathbf{r}_j - \mathbf{r}_i\|^3$ dependence; therefore, since we model our massive bodies as point particles, when two bodies collide or pass each other closely, the model becomes problematic and may become singular if both bodies exist at the same position at the same time step. The model may be improved by including appropriate collision handling, where the simulation is stopped when two bodies come within a predetermined distance from one another. Moreover, gravitational softening [5] may be implemented where we introduce a small positive value, ϵ , such that $1/r^3 \rightarrow 1/(r^2 + \epsilon^2)^{3/2}$. When r becomes small or equals zero, ϵ prevents the model from diverging or becoming singular. However, this removes the singularity at the cost of modifying the gravitational interaction at separations comparable to ϵ .

Furthermore, throughout this report, we advance our simulation in fixed time steps of a chosen Δt . During close encounters, the dynamic timescale can decrease rapidly; thus, a fixed time step that is adequate elsewhere may become too large to resolve the motion accurately. A solution to this issue is to include an adaptive integrator where the value of Δt changes depending on a local error estimate or relevant dynamical timescale. Since these methods directly calculate the gravitational acceleration at each time step for each body, the current implementation evaluates interactions directly between all bodies, given an $\mathcal{O}(N^2)$ force-calculation cost per time step. This may be improved upon by introducing tree methods where, for large N , approximate tree methods such as Barnes-Hut [6] can reduce this scaling to approximately $\mathcal{O}(N \log N)$. Lastly, our simulation only considers Newtonian gravity and disregards all relativistic effects, which may become important in strong gravitational fields or at relativistic velocities.

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